sshower: A simple parton shower program

ΕΘΝΙΚΟ ΚΑΙ ΚΑΠΟΔΙΣΤΡΙΑΚΟ ΠΑΝΕΠΙΣΤΗΜΙΟ ΑΘΗΝΩΝ
NATIONAL AND KAPODISTRIAN UNIVERSITY OF ATHENS

Στυλιανός-Χρήστος Σπυρουνάκος
Stylianos-Christos Spyrounakos
7110112300206

Final project for the postgraduate course: Computational Physics

September 30, 2024

Outline

Introduction

The "normal" procedure to having higher multiplicity final states is to include calculations of higher perturbative orders. There can appear, however, logarithms in terms of the perturbative series, which can become large enough to change the convergence behaviour. This way, matrix elements that include even higher order terms could have significant contributions that are not taken into account. When such logarithms are large, fixed order predictions are unreliable. A possible remedy is "resummation", a procces that tries to sum such problematic terms to all orders.

Note:

Of course, another problem with including higher order terms is the difficulty.

These logarithmic terms are of the form $\alpha^n L^m$, where α is the coupling parameter of the expansion and L is, broadly speaking, a logarithm of the ratio of two different scales. As an example, for a quantity O with a perturbative expansion, we could have:

$$O = c_0 + \alpha(c_{10} + c_{11}L + c_{12}L) + \alpha^2(c_{20} + c_{21}L + \ldots + c_{24}L^4) + \ldots$$

and the resummed series would be:

$$O_r = \exp(Lf_1(\alpha L) + f_2(\alpha L) + \ldots)(c_0^{'} + c_1^{'}\alpha + \ldots),$$

where f_i are functions of αL. The accuracy, then, depends on how many terms of the exponent are kept for computations. With only the first term we have leading logarithmic (LL) accuracy.

Note:

The type of logarithm is also important.

A parton shower is a way to perfom resummation, that can give event type information an be used for numerical simulations. In essence, the parton shower approximates higher order corrections, enhanced by large kinematic logarithms and generates high-multiplicity final states that can be converted to observable particles through hoadronization. The enhanced terms of interest are those that have to do with soft and collinear emissions. In the leading logarithmic picture a parton shower can be implemented by a series of 1 to 2 branchings.

Note:

The classification of logarithmic parton shower accuracy is complicated, because, while they may not explicitly include higher order corrections, they have some features that include higher order effects compared to same order analytic resummation techniques (such as momentum conservation, which is respected all over phase space in parton showers, while only in the strict collinear and soft limits in analytic resummation).

The parton shower is a Markov process, where a series of values for an evolution variable t and an energy/momentum sharing variable z (and particle information) are generated in each branching using only information generated by the previous branching. Each branching is characterized by the appropriate (LO) splitting kernel $P_{a \to bc}$, for the branching of particle a to particles b and c.

Sudakov form factor

We begin with the probability of a branching $a \to bc$:

$$\mathrm{d}^2\mathcal{P}_{a\to bc} = \frac{\alpha_S}{2\pi}\frac{\mathrm{d}t}{t}\mathrm{d}z\, P_{a\to bc}(z,t),$$

where:

- we are working in the collinear approximation (small p_T and mass)
- α_S is the strong coupling constant (but could be another if we are concerned with something other than QCD), which depends on both z and t
- the splitting kernel generally depends on both z and t, but for massless particles they are simply the DGLAP equation unregulated kernels

Note:

Showing that this is the probability is outside of our scope, but a "simple" way is to compare the cross section of producing $n+1$ particles vs n $\to$ universal functions modify it.

This probability is infinite in the collinear and soft limits (we will se the form of the kernels later). These real emission divergences would be cancelled by virtual corrections $\rightarrow$ inclusion through initarity $\rightarrow$ we impose a cut-off for t classifying emission as non-resolvable beyond this point $\rightarrow$ finite resolvable emissions.

To this end, we write the probability that a branches to b, c in $\mathrm{d}t$:

$$\mathrm{d}\mathcal{P}_{a\to bc} = \frac{\mathrm{d}t}{t} \int_{z_-(t)}^{z_+(t)} \mathrm{d}z' \frac{\alpha_S}{2\pi} P_{a\to bc}(z', t),$$

where the z limits are due to kinematic constraints and the cut-off (we avoid the need for regulation at $z = 0, 1$).

Including all possible branchings, the probability of a resolvable branching of a (meaning t must be above the cut-off) in $\mathrm{d}t$ is:

$$\mathrm{d}\mathcal{P}_a = \frac{\mathrm{d}t}{t} \sum_{b,c} \int_{z_-(t)}^{z_+(t)} \mathrm{d}z' \frac{\alpha_S}{2\pi} P_{a\to bc}(z', t)$$

and, by unitarity, the probability for no resolvable branching in $\mathrm{d}t$ is:

$$1 - \mathrm{d}\mathcal{P}_a = 1 - \frac{\mathrm{d}t}{t} \sum_{b,c} \int_{z_-(t)}^{z_+(t)} \mathrm{d}z' \frac{\alpha_S}{2\pi} P_{a\to bc}(z', t)$$

Dividing a range $[t, t'']$ ($t > \mathit{cutoff}$) in steps of Δt, multiplying the no (resolvable) branching probabilities and taking the limit $\Delta t \to 0$ we obtain the Sudakov form factor:

$$\Delta_a(t, t'') = \exp\left(-\int_{t''}^{t} \mathrm{d}\mathcal{P}_a\right)$$
$$= \exp\left(-\int_{t''}^{t} \frac{\mathrm{d}t'}{t'} \sum_{b,c} \int_{z_-(t')}^{z_+(t')} \mathrm{d}z' \frac{\alpha_S}{2\pi} P_{a\to bc}(z', t')\right),$$

which is the probability of no resolvable emission from a between t'' and t.

With that, the probability of a evolving from t'' to t without emissions and then branching at t is:

$$\Delta_a(t, t'')\mathrm{d}\mathcal{P}_a(t).$$

Algorithm

The previous expression tells us that to generate a series of branching information, starting at a scale t_i and a cut-off at t_c, we can implement the following algorithm (for a specific branching $a \to bc$):

1. compare $\Delta_{a\to bc}(t_c, t_i)$ with a random number $\mathcal{R} \in (0, 1)$
   - if $\Delta_{a\to bc}(t_c, t_i) < \mathcal{R}$ continue to 2
   - else end the evolution
2. solve $\Delta_{a\to bc}(t, t_i) = \mathcal{R}$ for the t of the next emission
3. solve $\int_{z_-}^{z} \mathrm{d}z' \frac{\alpha_S}{2\pi} P_{a\to bc}(z', t') = \mathcal{R}' \int_{z_-}^{z_+} \mathrm{d}z' \frac{\alpha_S}{2\pi} P_{a\to bc}(z', t')$ for the z of the emission, where $\mathcal{R}'$ another random number
4. set t_i to t and start again from 1

A few comments:

- For more possible branchings the solution is simply to generate a t for each and keep the branching with the largest value of t
- Solving for t in the previous algorithm entails, defining $\Gamma(t') \equiv \frac{1}{t'} \int_{z_-}^{z_+} \mathrm{d}z' \, \frac{\alpha_S}{2\pi} P_{a\to bc}(z', t')$, knowing its primitive $\tilde{\Gamma}$ and the inverse of the primitve $\tilde{\Gamma}^{-1}$, which can be highly non-trivial
- To go around this the algorithm can be performed by numerically integrating Sudakov form factors and constructing look-up tables but there is a better option, the Sudakov veto algorithm

The Sudakov veto algorithm involves overestimating the branching probability (thus underestimating Δ and overestimating the number of emissions) to be able to solve for t and then vetoing emissions with appropriate weights to recover the original Δ/probabilities.

Let $\hat{P}_{a\to bc}$, $\hat{\alpha}_S$ and $\hat{z}_+$, $\hat{z}_-$ be the overestimates. It is important that they do not depend on the t' variable of integration. A constant value for $\hat{\alpha}_S$ and $\hat{P}_{a\to bc} = \hat{P}_{a\to bc}(z')$ are the most appropriate.

Defining

$$\rho_{a\to bc}(\hat{z}_+, \hat{z}_-) \equiv \int_{\hat{z}_-}^{\hat{z}_+} \mathrm{d}z' \, \frac{\hat{\alpha}_S}{2\pi} \hat{P}_{a\to bc},$$

with a primitive $\tilde{\rho}_{a\to bc}(z)$ $(\to \rho_{a\to bc}(z, z') = \tilde{\rho}_{a\to bc}(z) - \tilde{\rho}_{a\to bc}(z'))$ and inverse of the primitive $\tilde{\rho}^{-1}_{a\to bc}(z)$ we can solve for t:

$$t = t_i \mathcal{R}^{1/\rho_{a\to bc}(\hat{z}_+, \hat{z}_-)}$$

and z:

$$z = \tilde{\rho}^{-1}_{a\to bc}(\tilde{\rho}_{a\to bc}(\hat{z}_-) + \mathcal{R}'[\tilde{\rho}_{a\to bc}(\hat{z}_+) - \tilde{\rho}_{a\to bc}(\hat{z}_-)])$$

Note:

Solving for t is not necessarily as easy as it seems, since the overestimated limits may themselves depend on t and thus $\tilde{\rho}_{a\to bc}(z)$ as well.

Now that we have the t and z of the branching we veto it with weights:

$$\frac{\alpha_S(z,t)}{\hat{\alpha}_S} \text{ and } \frac{P_{a\to bc}(z,t)}{\hat{P}_{a\to bc}(z)},$$

$\to$ if $\frac{\alpha_S(z,t)}{\hat{\alpha}_S} < \mathcal{R}''$ or $\frac{P_{a\to bc}(z,t)}{\hat{P}_{a\to bc}(z)} < \mathcal{R}'''$ ($\mathcal{R}''$, $\mathcal{R}'''$ random numbers) we reject the branching. We also veto if the p_T^2 of the emission is unphysical (< 0). If the branching is rejected the process is repeated after setting $t_i = t$.

Lastly, an azimuthial angle for the emission also needs to be generated. The simplest way is to generate a uniformly random angle (this is what we will do), but for the complete picture spin correlations and gluon polarization need to be taken into account.

Kinematics

We handle the kinematics of a branching as simply as possible: Particle a just branches and we assign momenta to the resulting particles b, c such that momentum is conserved (using the z and p_T that the emission generated). We consider all particles massless.

More sophisticated (and correct) algorithms would track color connections between the particles and assign a recoil partner to the emitting particle so that it can branch (the emitting particle must be off-shell).

After the shower is over there is a need to reconstruct the kinematics to ensure momentum is conserved globally, since the particles that get showered go off their mass shell to branch and the final particles are set on-shell.

Before the shower:

The momenta of the progenitor partons (meaning the partons of the hard process that will be showered) are $p_J = \left(\sqrt{\vec{p_J}^2 + m_J^2}, \vec{p_J}\right)$, where $J = 1, \ldots, n$ and

- $\sum_{J=1}^{n} \sqrt{\vec{p_J}^2 + m_J^2} = \sqrt{s}$
- $\sum_{J=1}^{n} \vec{p_J} = \vec{0}$

After the shower:

The momenta of the jets (jet meaning here the particles of the shower of a progenitor) are $q_J = \left(\sqrt{\vec{q_J}^2 + q_J^2}, \vec{q_J}\right)$

The easiest and less intrusive way to conserve momentum is to

1. rotate each jet to the direction of its progenitor
2. boost the momenta of the jet particles by rescaling: $\vec{q_J}^2 \to k^2 \vec{p_J}^2$, such that $\sum_{J=1}^{n} \sqrt{k^2 \vec{p_J}^2 + q_J^2} = \sqrt{s}$

Specifics and Implementation

Choice of evolution variable

As the evolution variable (massless) we will use $\tilde{q}^2 = E_a^2 \frac{p_b p_c}{E_b E_c}$, which captures coherence effects and leads to angular ordered emissions and z will be the energy fraction $z = \frac{E_b}{E_a}$. With that:

- the angular ordering condition for the next emission scales is $\tilde{q}_b^2 < z^2 \tilde{q}^2$, $\tilde{q}_c^2 < (1-z)^2 \tilde{q}^2$
- the cutoff on the emission scale is $\tilde{q}_{cut}^2 = 4\tilde{q}_0^2$, where $\tilde{q}_0$ is the cut-off scale for the shower
- $z_+ = 1 - \frac{\tilde{q}_0}{\tilde{q}}$, $z_- = \frac{\tilde{q}_0}{\tilde{q}}$ (these will also be used as "overestimates" as they do not depend on the integration variable of the Sudakov form factor)
- the transverse momentum of the emission is $p_T^2 = z^2(1-z)^2\tilde{q}^2$
- the virtual mass the emitting particle obtains is $m_a^2 = z(1-z)\tilde{q}^2$

Scales of α_S

- The scale of the coupling constant for each emission will be the p_T^2 of the emission to capture some higher order effects
- The overestimate of the coupling constant will be its value at the cut-off scale of the shower

Choice of initial and cut-off scales

- The cut-off scale needs to keep the shower evolution from going to non-perturbative territory at around $1\,\mathrm{GeV}$, when hadronization models take over. We will set $\tilde{q}_0 = 0.935\,\mathrm{GeV}$ (agrees with the Herwig 7 choice)
- The easiest and most symmetric choice for the starting scale is the center-of-mass energy

Splitting kernels and related functions (massless)

- For $q \to gq$:
  - $P_{q\to gq} = C_F \frac{1+z^2}{1-z}$
  - $\hat{P}_{q\to gq} = C_F \frac{2}{1-z}$
  - $\tilde{\rho}_{q\to gq} = -2C_F \frac{\hat{\alpha}_S}{2\pi} \ln(1-z)$
  - $\tilde{\rho}^{-1}_{q\to gq} = 1 - \exp\left(\frac{z}{2C_F \frac{\hat{\alpha}_S}{2\pi}}\right)$
- For $g \to q\bar{q}$:
  - $P_{g\to q\bar{q}} = T_R[1 - 2z(1-z)]$
  - $\hat{P}_{g\to q\bar{q}} = T_R$
  - $\tilde{\rho}_{g\to q\bar{q}} = T_R \frac{\hat{\alpha}_S}{2\pi} z$
  - $\tilde{\rho}^{-1}_{g\to q\bar{q}} = \frac{z}{T_R \frac{\hat{\alpha}_S}{2\pi}}$
- For $g \to gg$:
  - $P_{g\to gg} =$ $C_A \left[\frac{z}{1-z} + \frac{1-z}{z} + z(1-z)\right]$
  - $\hat{P}_{g\to gg} = C_A \left[\frac{z}{1-z} + \frac{1-z}{z}\right]$
  - $\tilde{\rho}_{g\to gg} = -C_A \frac{\hat{\alpha}_S}{2\pi} \ln\left(\frac{1}{z} - 1\right)$
  - $\tilde{\rho}^{-1}_{g\to gg} = \frac{1}{1+\exp\left(-\frac{z}{C_A \frac{\hat{\alpha}_S}{2\pi}}\right)}$

Implementation specifics

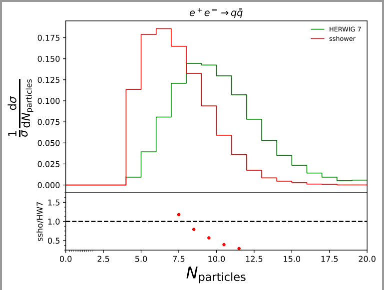
The code can be found at
`https://github.com/stelios-spyrounakos/sshower.git` and is made to shower $e^+e^- \to q\bar{q}$ events in the sense that some checks in the code and the io assume that only these particles exist and it is an easy enough case so that this sipmle code works adequately. The input is a .lhe(.gz) file and it outputs a .lhe file

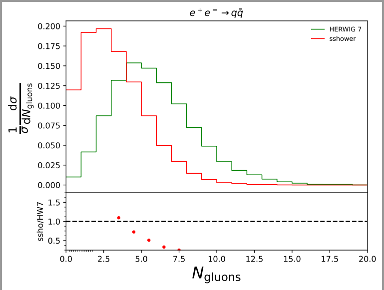
- The value of $\tilde{q}^2$ for a branching is found numerically, solving $\ln(\frac{\tilde{q}^2}{\tilde{q}_i^2}) - \frac{1}{\rho_{a\to bc}(\hat{z}_+, \hat{z}_-)} \ln \mathcal{R} = 0$
- Branchings take place in the direction of the z axis and then the children parton are rotated to the original direction of the parent parton
- The k factor for the boosts is easily computed analytically for 2 jets under the assumption that the hard process conserves momentum: $k^2 = \frac{(s-q_1^2-q_2^2)^2-4q_1^2q_2^2}{4sp_1^2}$

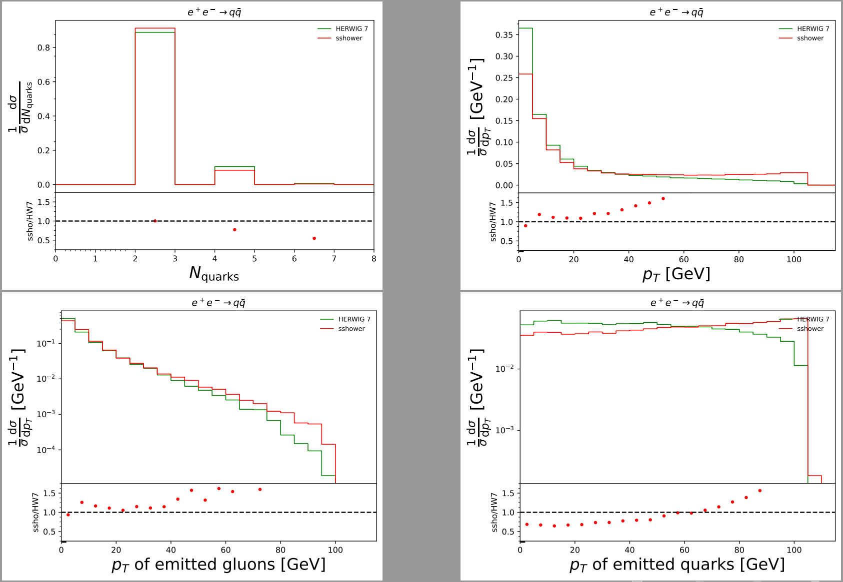
- The β factor for the boost is easily computed under the assumption that $\vec{q} \parallel \vec{p} \parallel \vec{\beta}$: $|\vec{\beta}| = \frac{|\vec{q}|\sqrt{\vec{q}^2+q^2} - k|\vec{p}|\sqrt{k^2\vec{p}^2+q^2}}{k^2\vec{p}^2+\vec{q}^2+q^2}$
- C++11 standard and newer for compilation
- The GSL (GNU Scientific Library) was used for numerical root finding
- The Eigen library was used for linear algebra
- zlib was used for handling compressed input files

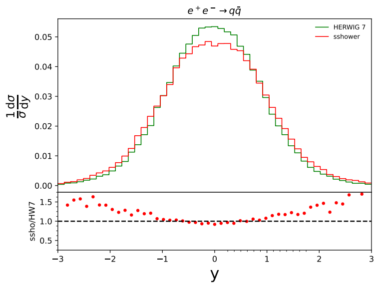
Comparison plots

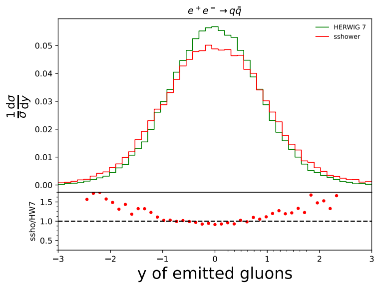
We compare the results of sshower with the corresponding results of showering the same initial sample of $e^+e^- \to q\bar{q}$ (produced by MadGraph) using Herwig 7.
The fastjet library was used for jet clustering(anti-kt algorithm with $R = 0.4$)

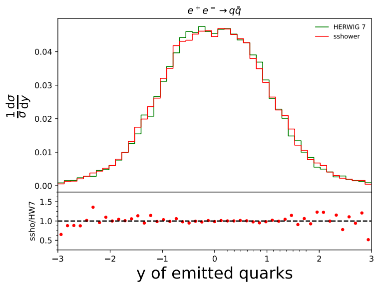


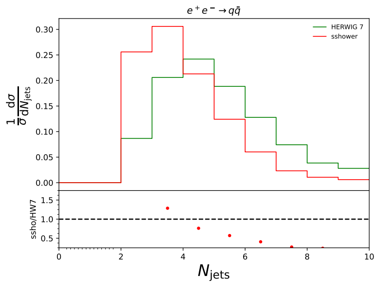


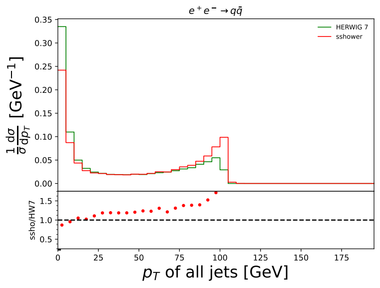

$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dN_{quarks}}$
ssho/HW7
N_{quarks}
$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dp_T}$ [GeV^{-1}]
ssho/HW7
p_T [GeV]
$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dp_T}$ [GeV^{-1}]
ssho/HW7
p_T of emitted gluons [GeV]
$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dp_T}$ [GeV^{-1}]
ssho/HW7
p_T of emitted quarks [GeV]

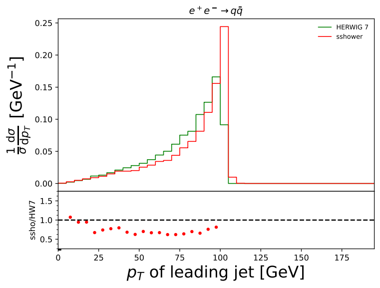

$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dy}$
ssho/HW7
y

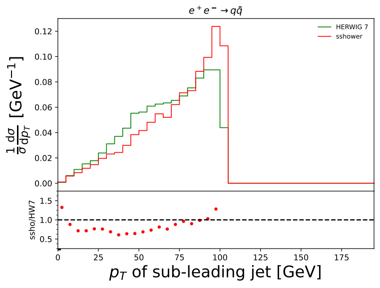

$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dy}$
ssho/HW7
y of emitted gluons

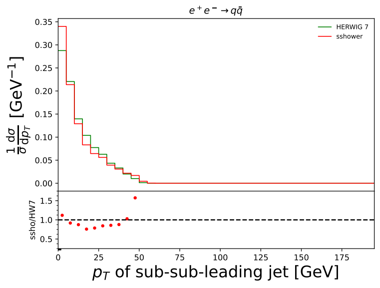

$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dy}$
ssho/HW7
y of emitted quarks

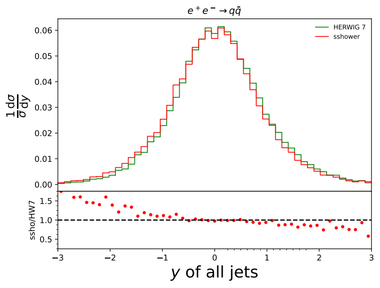

$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dN_{jets}}$
ssho/HW7
N_{jets}

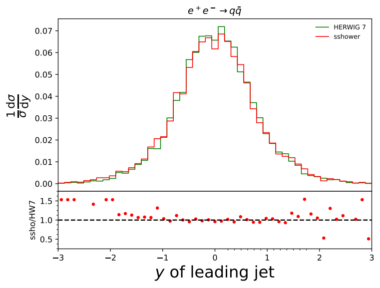

$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dp_T}$ [GeV^{-1}]
ssho/HW7
p_T of all jets [GeV]

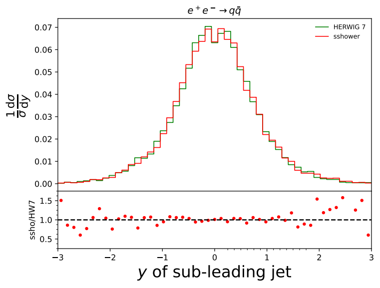

$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dp_T}$ [GeV^{-1}]
ssho/HW7
p_T of leading jet [GeV]

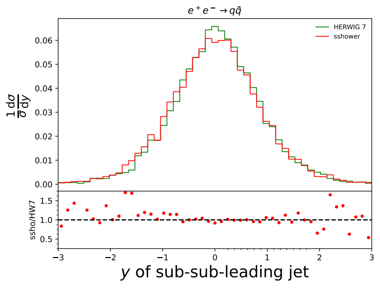

$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dp_T}$ [GeV^{-1}]
ssho/HW7
p_T of sub-leading jet [GeV]

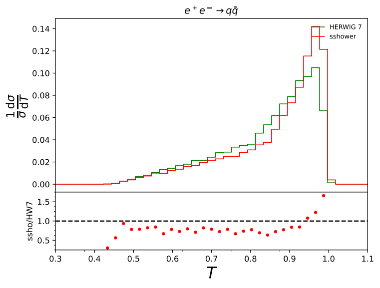

$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dp_T}$ [GeV^{-1}]
ssho/HW7
p_T of sub-sub-leading jet [GeV]


$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dy}$
ssho/HW7
y of all jets


$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dy}$
ssho/HW7
y of leading jet


$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dy}$
ssho/HW7
y of sub-leading jet


$e^+e^- \to q\bar{q}$
HERWIG 7
sshower
$\frac{1}{\sigma}\frac{d\sigma}{dy}$
ssho/HW7
y of sub-sub-leading jet



To make Herwig as close to sshower in its simplicity as possible:

- Of course, only the same types of splittings are turned on
- Hadronization is turned off
- Spin correlations are turned off

Final comments

Even though we have limited herwig, the results are actually pretty good considering that Herwig:

- is a mature, way more advanced and sophisticated program, that is used for research and not merely for demonstration purposes
- the evolution variable, as well as the splitting kernels take mass into account
- handles the (massive) kinematics and the reconstruction better
- tracks color flow

Note:

Actually, $g \to q\bar{q}$ emissions should not be subject to angular ordering and they indeed aren't in Herwig 7.


That's all Folks!

Bibliography

- Papaefstathiou, Andreas (2024). *Pyresias: How To Write a Toy Parton Shower*. arXiv: 2406.03528 [hep-ph]. URL: https://arxiv.org/abs/2406.03528.
- Bähr, Manuel et al. (2008). "Herwig++ physics and manual". In: *The European Physical Journal C* 58.4, pp. 639–707. ISSN: 1434-6052. DOI: 10.1140/epjc/s10052-008-0798-9. URL: http://dx.doi.org/10.1140/epjc/s10052-008-0798-9.
- Gieseke, Stefan, Alberto Ribon, et al. (2004). "Herwig++ 1.0: an event generator for e+e- annihilation". In: *Journal of High Energy Physics* 2004.02, p. 005. DOI: 10.1088/1126-6708/2004/02/005. URL: https://dx.doi.org/10.1088/1126-6708/2004/02/005.
- Ellis, R. K., W. J. Stirling, and B. R. Webber (1996). *QCD and Collider Physics*. Cambridge Monographs on Particle Physics, Nuclear Physics and Cosmology. Cambridge University Press.
- Campbell, John, Joey Huston, and Frank Krauss (2017). *The Black Book of Quantum Chromodynamics: A Primer for the LHC Era*. Oxford University Press. ISBN: 9780199652747. DOI: 10.1093/oso/9780199652747.001.0001. URL: https://doi.org/10.1093/oso/9780199652747.001.0001.

- Bellm, Johannes et al. (2016). “Herwig 7.0/Herwig++ 3.0 release note”. In: *The European Physical Journal C* 76.4. ISSN: 1434-6052. DOI: 10.1140/epjc/s10052-016-4018-8. URL: http://dx.doi.org/10.1140/epjc/s10052-016-4018-8.
- Sjöstrand, Torbjörn, Stephen Mrenna, and Peter Skands (2006). “PYTHIA 6.4 physics and manual”. In: *Journal of High Energy Physics* 2006.05, pp. 026–026. ISSN: 1029-8479. DOI: 10.1088/1126-6708/2006/05/026. URL: http://dx.doi.org/10.1088/1126-6708/2006/05/026.
- Bierlich, Christian et al. (2022). *A comprehensive guide to the physics and usage of PYTHIA 8.3*. arXiv: 2203.11601 [hep-ph]. URL: https://arxiv.org/abs/2203.11601.
- Dobbs, M. A. et al. (2004). *Les Houches Guidebook to Monte Carlo Generators for Hadron Collider Physics*. arXiv: hep-ph/0403045 [hep-ph]. URL: https://arxiv.org/abs/hep-ph/0403045.
- Campbell, J. M. et al. (2024). *Event Generators for High-Energy Physics Experiments*. arXiv: 2203.11110 [hep-ph]. URL: https://arxiv.org/abs/2203.11110.

Höche, Stefan (2015). *Introduction to parton-shower event generators*. arXiv: 1411.4085 [hep-ph]. URL: https://arxiv.org/abs/1411.4085.

Gieseke, Stefan, Philip Stephens, and Bryan Webber (2003). "New formalism for QCD parton showers". In: *Journal of High Energy Physics* 2003.12, pp. 045–045. ISSN: 1029-8479. DOI: 10.1088/1126-6708/2003/12/045. URL: http://dx.doi.org/10.1088/1126-6708/2003/12/045.

Marchesini, G. and B. R. Webber (1988). "Monte Carlo simulation of general hard processes with coherent QCD radiation". In: *Nuclear Physics B* 310.3, pp. 461–526. ISSN: 0550-3213. DOI: 10.1016/0550-3213(88)90089-2. URL: https://www.sciencedirect.com/science/article/pii/0550321388900892.

Marchesini, G. and B. R. Webber (1984). "Simulation of QCD jets including soft gluon interference". In: *Nuclear Physics B* 238.1, pp. 1–29. ISSN: 0550-3213. DOI: 10.1016/0550-3213(84)90463-2. URL: https://www.sciencedirect.com/science/article/pii/0550321384904632.

Buckley, Andy et al. (2011). "General-purpose event generators for LHC physics". In: *Physics Reports* 504.5, pp. 145–233. ISSN: 0370-1573. DOI: 10.1016/j.physrep.2011.03.005. URL: http://dx.doi.org/10.1016/j.physrep.2011.03.005.